

Scenario	Details
Cloud in a box	Local caching of content or providing a whole stack of compute, network, storage in a self-contained fashion near the usage site overcomes the limitation of networking.
Mobile connectivity	Ubiquitous use of mobiles, specially with the emergence of 5G, is going to benefit from computing moving closer to the workloads.
Network-as-a-service	Small footprint of network service provider's distribution network at the edge makes it more affordable.
Universal Customer Premises Equipment (uCPE)	Applicable especially for NFV, where different sites need to host different applications, but they need to work in concert.
Satellite enabled communication (SATCOM)	Massive distribution of terminal devices often in very remote and adverse conditions, e.g., aircrafts, oil exploration, mining, and military grade instruments.

Table 3: Cloud edge deployment scenarios

indicators (KPIs), optimal resource utilisation and the quality of service (QoS). There are various well-studied techniques that are available, namely but not limited to:

- Linear programming based resource placement
- Graph to graph mapping of resources
- Multi-objective optimisations

Resource elasticity and remediation: The following techniques are applicable here:

1. Geographical load balancing through cost function modelling, considering service delay and energy consumption.
2. Placement decision based on attributes of every single container (size and resource demand).
3. Particle swarm optimisation algorithm (multiple objective particle swarm optimisation, MOPSO).

Network function virtualisation: Network Function Virtualisation (NFV) is a technique for modelling the network functions into basic building blocks at the software layer.

The primary advantage is that these NFVs can run on any standard server or cloud without requiring any specialised network appliance. Typical techniques of NFV resource allocations are:

1. Markov decision process.
2. Graph neural network model based prediction. The training is done with historical and topological information (i.e., behaviour of neighbouring components).

3. Often, genetic programming approaches are used, as they do better than their linear counterparts.

Open source in cloud-edge

This is really an emerging field and tools are yet to show up. However, OpenStack – which is a widely used cloud data management framework – is also delving into this field through its initiative called OpenStack Foundation Edge Computing Working Group (OSF Edge WG). It has classified the scenarios into various categories, as shown in Table 3.

Going forward

Resource provisioning in cloud deployment is a well-studied subject. We have discussed the various tenets associated with the problem statement, which leads to a general reference architecture consisting of various modellers, predictors and optimisers to get optimal placement decisions for placement planning.

We also discussed some of the underlying techniques to get things done in a viable and feasible way. We hope that the overall architectural reference diagram along with the techniques in our arsenal will be invaluable for further studies on the resource provisioning problem. **END** 

References

- [1] Machine Learning Methods for Reliable Resource Provisioning in Edge-Cloud Computing: A Survey, <https://dl.acm.org/doi/10.1145/3341145>
- [2] Mobile-Edge Computing and the Internet of Things for Consumers: Extending cloud computing and services to the edge of the network, <https://ieeexplore.ieee.org/abstract/document/7574425>
- [3] IBM. 2013. Smarter wireless networks, https://www.ibm.com/services/multimedia/Smarter_wireless_networks.pdf.
- [4] S.N. Shirazi, A. Gouglidis, A. Farshad, and D. Hutchison. 2017. The extended cloud: Review and analysis of mobile edge computing and fog from a security and resilience perspective. IEEE J. Select. Areas Commun. 35, 11 (Oct. 2017), 2586–2595.
- [5] Cloud Edge Computing: Beyond the Data Center, <https://www.openstack.org/use-cases/edge-computing/cloud-edge-computing-beyond-the-data-center/>

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